Data-Driven Analysis for the Operation Status of the e-Commerce Platform Based on Olist

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Abstract: Based on the Olist sales data and user feedback from 2017 to 2018, this paper studies the sales situation of Olist, and discusses how enterprises can further stimulate sales, strengthen cooperation between the platform and merchants, and optimize the customer experience. The main method of this study is to use Python to visualize the data and to conduct statistical analysis on the data. Through the analysis, we find that there is a lack of customer loyalty for Olist. Platform services, cooperation with suppliers and product quality control also should be improved. The products and services provided by the platform have not reached the level of consumer satisfaction. The time difference between the platform and the supplier also exists. In this paper, specific suggestions are given to solve the existing problems of Olist. These suggestions can also be applied to other e-commerce platforms with similar operating conditions as Olist to increase sales and customer satisfaction.

1 INTRODUCTION

With the development of business, statistics on business results have become a very important part of business activities. Any business can gain much profit through scientific data analysis. For example, enterprises can find various problems in the development of enterprises. For managers, the scary thing is not the emergence of a problem or crisis, but the lack of a solution or control the various data of products within a reasonable range, understand the relationship between supply and demand, know which products are popular and which products are not profitable, and adjust business ideas in time to allow enterprises to obtain more valuable information. Business analysis and data science can be combined.

Although there are several definitions for the concept of business analysis, it is difficult to define it as a set of numerical tools, methodology, business processes, and analytical modeling methods that allow data-driven decision-making in modern companies and organizations. To this end, it makes extensive use of data and information science.

In this case, we collected some data from Olist, which is the largest department store in Brazilian marketplaces. Olist connects small businesses from all over Brazil to channels without hassle and with a single contract. Since we have seen the boss's needs to process daily operational data on the Internet, we are ready to use our capabilities to solve these problems, and help them to make final decisions.

With the e-commerce platform's development, many pieces of literature studied different business models. There are many factors that affect the customer's purchase. Kwahk and Kim (Kwahk & Kim 2017) analyze results show that social interaction ties have significant positive effects on social impact transfer factors and trust in online vendors, whereas they do not directly influence visit intention. Social media commitment plays a crucial role in increasing social impact transfer factors and e-commerce outcomes. Flanagin et al (Flanagin, Metzger, Pure, Markov& Hartsell 2014) suggest that despite valuing the web and ratings as sources of commercial information, people use rating information suboptimally by potentially privileging small numbers of ratings that could be idiosyncratic. And the product

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quality is shown to mediate the relationship between user ratings and purchase intention. Buttle et al. (Buttle 1997) carried out mail surveys on 4,250 certificated organizations; 1,220 (28.7 percent) responded. Marketing considerations were secondary in seeking registration, and outcomes related to profitability and process improvement were more highly valued than marketing benefits.

Logistics and transportation are also important for e-commerce companies. Speranza et al. (Speranza 2018) propose that problems in transportation and logistics had to be tackled long before computers and Operational Research (OR) became available to support decision making. After the first optimization models were developed, OR has substantially contributed to making transportation systems efficient and companies with complex transportation and logistics problems competitive. Gunasekaran and Ngai et al. (Gunasekaran & Ngai 2003) analyze the company, a small third-party logistics (3PL) company in Hong Kong, which has been successful in its overall business performance and satisfying customers. This company's strategic alliances with both clients and customers have helped to improve the utilization of its resources, such as warehouse space and transportation fleets. Also, the company is in the process of expanding its operations across greater China, to become a full-fledged 3PL company.

E-commerce is becoming more popular. Sarkis and Talluri (Sarkis, Meade & Talluri 2004) realize that a strong supporting logistics or electronic logistics (e-logistics) function is an important organizational offering from both the commercial and the consumer perspective. The implications of e-logistics models and practices cover the forward and reverse logistics functions of organizations. They also have a direct and profound impact from an environmental perspective. Barnes et al. (Barnes 2002) analyze that fueled by the increasing saturation of mobile technology, such as phones and personal digital assistants (PDAs), m-commerce promises to inject considerable change into the way certain activities are conducted. Equipped with micro-browsers and other mobile applications, the new range of mobile technologies offers the Internet 'in your pocket' for which the consumer possibilities are endless, including banking, booking or buying tickets, shopping, and real-time news. Bichler and Zhao (Bichler, Segev & Zhao 1998) component-based e-commerce technology is a recent trend towards resolving the e-commerce challenge at both system and application levels. Instead of delivering a system as a prepacked monolith system containing any conceivable feature, component-based systems consist of a lightweight kernel to which new features can be added in the form of components.

Jing Dong is one of the giants in the Chinese market. Li Mei and Guo Chen (Li & Guo 2015) research the issue of self- logistics in Jing Dong under the background of the rapid development of e-commerce. The analysis of the advantages and disadvantages and the comparison of the third-party logistics proved that self- logistics enjoyed significant advantages. However, there exist potential risks of too large investment, difficult management of staff, and distraction from core business. Sun and Zhou (Sun, Liu, Higgs & Zhou 2017) consider the research status of the Internet of Things as applied to supply chain management. A supply chain analysis model is built under the Internet of Things environment using Jing Dong Mall as a case study. The results show that the Internet of Things improved the quality of information available to Jing Dong, enhanced its management efficiency, and improved customers' satisfaction; it also reduced the cost of supply chain management whilst creating more new value.

In this paper, we mainly analyze the e-commerce orders of Olist Store in Brazil from 2017 to 2018 focused on four dimensions: monthly sales, order confirmation time, scores of reviews, and the number of regular customers.

Firstly, we can see that sale volume has been increasing throughout 2017 and reached the highest level of the year in November. The decline was significant in December. Sales have declined every month since 2018, but overall sales are higher than 2017. Due to the last few months of 2018 being missing, we do not know how the sales are going in the rest of the months. Therefore, we guess it will hit a new high. Secondly, we analyze the order time. We know that the confirmation time for each quarter was less than 5 minutes, but similar to the previous analysis, due to the autumn and winter demand being high, the confirmation time for spring and summer was higher than autumn and winter. Thirdly, due to the user evaluation, it is one of the indicators used to evaluate the quality of goods, so we analyze the user evaluation. 59% of users gave a 5 score and 13% gave a 1-2 score. Overall, customer satisfaction at Olist stores is not very high. As a result, the total number of user reviews dropped significantly in August 2017. We speculate that the reason influencing this factor is product quality or platform service. Because the repurchase rate is the key to the sustainable operation of Olist stores. In the end, we analyze the number of regular customers. 90% of customers buy once a month, and less than 13% of customers buy more than three times a month. For this purpose, the Olist Store

needs to think about how to increase the short-time repurchase rate.

By analyzing we discuss the following ways: (1) We need to improve the efficiency of order confirmation in spring and summer; (2) it is suggested that Olist needs to strengthen its quality control; (3) it is suggested that Olist may collect customer's email and other information, and hold activities such as recommending and promoting products.

In this article, we first explained the background of the data. we will first conduct business analysis, and then combine customer data to analyze customer behavior on the basis of business analysis. Finally, we will summarize all the above analyses, draw our own conclusions and corresponding recommendations.

2 DATA BACKGROUND

In this paper, we used the data as a Brazilian e-commerce public dataset of orders made at Olist Store. The dataset has information of 100k orders from 2017 to 2018 made at multiple marketplaces in Brazil. Its features allow viewing order from multiple dimensions: from order status, price, payment, and freight performance to customer location, product attributes and finally reviews written by customers. We also released a geography dataset that relates Brazilian zip codes to let/lng coordinates. Next, we mainly study four aspects, such as monthly sales, order confirmation time, scores of reviews, and the number of regular customers.

In this picture, we can see that there are four variables that affect orders. These four factors are monthly sales, order confirmation time, scores of reviews, and the number of regular customers. They restrict each other. And these factors also have their own restricting factors. The reason for affecting order confirmation time is a customer's location, freight performance and order statues. Monthly sales and the number of regular customers have the same restricted factors. It is about price and product attributes. Scores of reviews have only affected reasons for reviews written by customers.

3 BUSINESS ANALYSIS

In this section, we selected some data from the homepage of Olist. After that, we used these data to do sales analysis, delivery time analysis and order time analysis.

3.1 Sales Analysis

We first created a table to look at the total sales of each month. We found that each month's sales were increasing. However, due to the missing data in the last few months of 2018, we were unable to make a judgment. As we can see from the Table 1, the maximum sale is in November, which is 7544.Then we take a graph for the sales amount from 2017 to 2018.

From the total sales (Figure 1), it can be seen that regardless of the lack of data, the sales volume showed

Figure 1: Variable diagram.

Table 1: Total sales for each month in 2018 and 2017.

Month	Jan	Feb	Mar	Apr	May	Jun
2017	800	1780	2682	2404	3700	3245
2018	7269	6728	7211	6939	6873	6167

Month	Jul	Aug	Sep	Oct	Nov	Dec
2017	4028	4331	4285	4631	7544	5673
2018	6292	6512	/	/	/	/

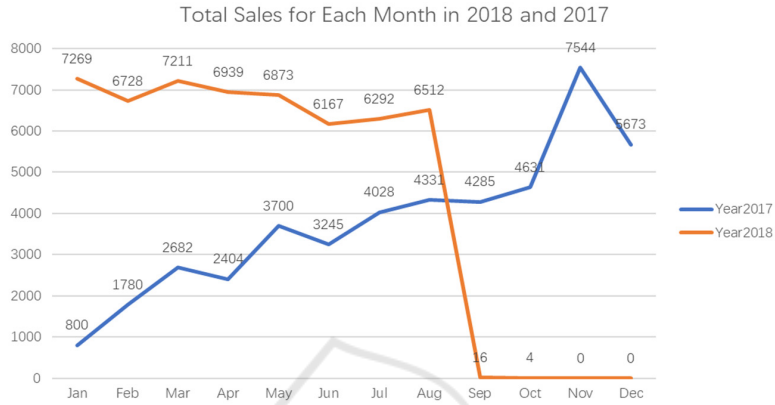

Figure 2: Total sales from 2017 to 2018 sales.

Table 2: Delivery Time Information (Excerpt).

	Customer date	Delivery date	Season
1	2017-10-10	2017-10-18	autumn
2	2018-08-07	2018-08-13	summer
3	2018-08-17	2018-09-04	summer
4	2017-12-02	2017-12-15	autumn
5	2018-02-16	2018-02-26	winter

	Time difference	Difference float
1	8days+21:25:13	-7.107488
2	6days+15:27:45	-5.355729
3	18days+18:06:29	-17.245498
4	13days+00:28:42	-12.980069
5	10days+18:17:02	-9.238171

a rapid upward trend in 2017 and reached its peak in November. After that, it has remained stable and fluctuates at around 6,500. It can be seen from this that 2017 was the peak period of Olist's development, and the order volume increased sharply. By 2018, the sales volume stabilized and the development entered the stabilizer. Data to speculate how to design marketing strategies.

3.2 Delivery Time Analysis

We counted the logistics speed of each season. The method of analysis is to make the difference between the estimated delivery date provided by the store and the actual delivery date, and finally get the difference date. The time difference column is obtained by subtracting the second column (delivery date) from the first column (customer date), the last column (difference float) is the number that turns the date into decimal Part of the whole table is shown in table 2.

Afterward, we plotted the difference (Figure 2) in days in different seasons and found that no matter what season, the customer's delivery time was much earlier than the expected delivery time, about a week earlier on average. According to the distribution of the histogram, the logistics rate in autumn and winter is significantly higher than that in spring and summer. The reason may be that the demand in spring and summer is larger, while the demand in autumn and winter is less.

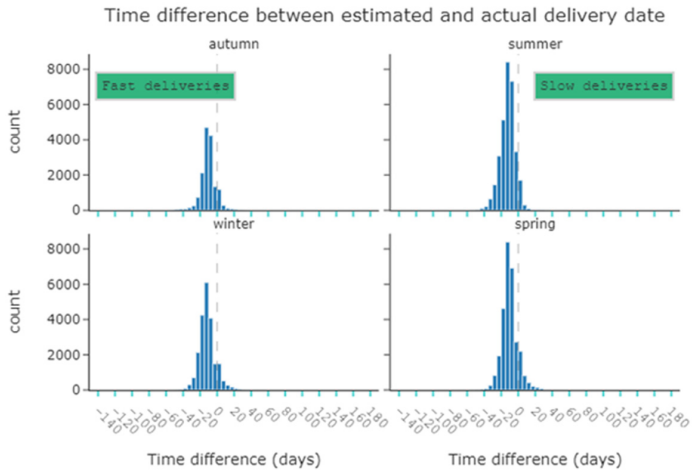

Figure 3: Time difference between estimated and actual delivery time.

Table 3: Order Time Information (Excerpt).

	Order purchase timestamp	Order approved at
1	2017-10-02 10:56:33	2017-10-02 11:07:15
2	2018-07-24 20:41:37	2018-07-26 03:24:27
3	2018-08-08 08:38:49	2018-08-08 08:55:23
4	2017-11-18 19:28:06	2017-11-18 19:45:59
5	2018-02-13 21:18:39	2018-02-13 22:20:29

	season	Time difference	Time difference float
1	autumn	0 days 00:10:42	0.007431
2	summer	1 days 06:42:50	1.279745
3	summer	0 days 00:16:34	0.011505
4	autumn	0 days 00:17:53	0.012419
5	winter	0 days 01:01:50	0.042940

3.3 Order Time Analysis

We then compared the difference between the payment time and the merchant's approval time in different seasons. The time difference column is obtained by subtracting the second column (delivery date) from the first column (customer date), the last column (time difference float) is the number that turns the date into decimal.

According to Figure 4, we found that the approval time for each season is within five minutes. Similar to the project analyzed in the previous analysis, the approval time for the spring and summer seasons is higher than that for the autumn and winter seasons. The reason may also be due to the large number of orders in the spring and summer seasons. In the autumn and winter seasons, more than 80% of the response time was within one day. In the spring and summer seasons, due to the large sample size and network delays caused by many orders, the response time within one day dropped to 70%.

Through the above analysis, we can summarize that it can be seen from the sales volume that the overall Olist is still developing, but the speed of development is slowing down. This is a problem worth thinking about, and a strategy needs to be devised to solve this problem. Turning our attention to logistics and response time, it can be seen that although the overall is still within an acceptable time range, there is still a lot of room for improvement, because, with the development of the times, people's tolerance for waiting time will decline, so a method to increase the speed of logistics is very necessary.

4 CUSTOMER'S ANALYSIS

In this part, through the analysis of customer's review scores and the change of customers repurchases. We can figure out how the public praise of Olist's products changed and give Olist the corresponding future development plans.

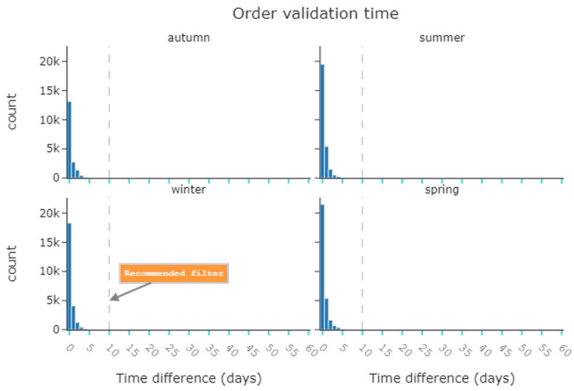

Figure 4: Order Validation Time.

Table 4: Customer's Review Scores.

	month	score	Review numbers
1	January	1-2	1032
2	January	3-4	1806
3	January	5	3695
4	February	1-2	1170
5	February	3-4	2114

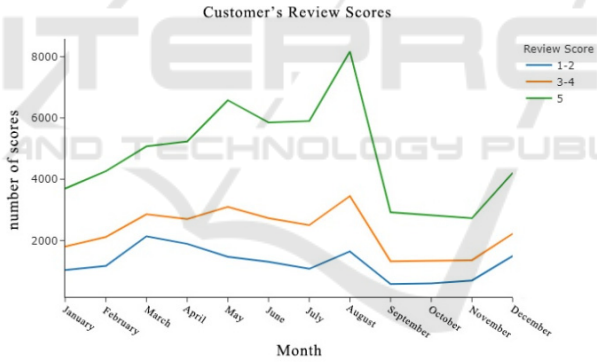

Figure 5: Customer's Review Scores.

4.1 Score Evaluation

In order to observe customers' response to the project, according to the Olist's data source, we counted the number of costumer's review scores and made a corresponding table. In the Table 4, we can find that no matter what month it is, the number of customers who give 5 points is the largest, then followed by customers who give 3 to 4 points. And customers who give 1 to 2 scores are the least.

It is widely known that the scores given by customers have a great relationship with the quality of goods and clearly reflects the feelings of customers after they purchase commodities. Besides, users' review rating is the most critical index to evaluate the quality of goods and services provided by e-commerce platforms.

We can see clearly in the Figure 5 that Olist's overall customer satisfaction is not satisfactory. Only 59% of customers gave a rating of 5 points out of 5 points. 13% of users gave negative reviews of one or two points. In the first eight months, we can see a steady change in the data. In August, the number of five-point positive reviews suddenly dropped sharply. Then it became smooth after September and then steadily increased again after November.

In August, there was an unusual drop in users' ratings. Such an abnormal situation may be due to

large-scale quality problems of the products being sold, or serious service errors of the platform. Considering the abnormal decline of both sales and reviews in August, it is suggested that Olist should strengthen its quality control. For Olist, ensuring the quality of products should be the top priority. Only by improving the quality of goods can customers' satisfaction after purchase be increased.

4.2 Regular Customers

At last, we counted the number of the customers' repurchases. Attracting customers to spend repeatedly on the platform is the key point of the sustainable business of e-commerce because whether it can attract customers to repurchase is also an important indicator of the electronic commercial success.

In order to observe the changes of the data more easily, we still make the data into a line chart. In the chart, the blue line means customer buy this product for the first time, the orange line means customer buy it the second time and the green line means customer purchases three or more times.

According to the Figure 6, we can draw the conclusion that before August, the volume of customers' first purchase increased steadily at a slow and steady pace. However, in August there was a sharp decline due to some specific reasons. After September, there has been a significant rebound in the amount of data of customer purchase. Meanwhile, we can find a problem that during the whole period of the year, the number of second and third purchases remained steadily at a lower level above 1,000, without drastic changes, and even did not change with the data of first purchases.

According to the data of regular customers, Olist had a bad performance in getting customers to make orders again. In 2017, more than 90% of Olist's users placed only one order per month, and less than 3% of customers spent three or more orders per month on Olist. Olist needs to think about how to motivate users to make multiple purchases in a short period of time. According to the marketing strategy, it is suggested that Olist may collect customers' email and other information, recommend products and promotion activities regularly, and develop different activities according to different themes throughout the year, so as to attract customers to repeat shopping on the platform within a short period of time.

By comparing the number of customer's first purchases and the scores given by customers. We can find that they were inextricably linked. The trend was roughly the same between them both. It suggesting that there is a huge connection between the customers' purchase and the public praise of the products.

According to the above conclusion, we suggest that Olist should increase the logistics speed in the spring and summer seasons because this is greatly related to customer satisfaction and the time from order to confirmation can be maintained cause this time is within an acceptable range. Otherwise, Olist is not doing well in terms of the repeat customers, they should take appropriate actions to recall regular customers, such as discounts on products and so on, to attract customers to repeat purchases. It also can be seen from the discount chart that the quality of products in August and September may decline, which leads to a decline in repurchases. It is necessary to strengthen the quality control to make customers most satisfied. And the public praise of products will also improve accordingly.

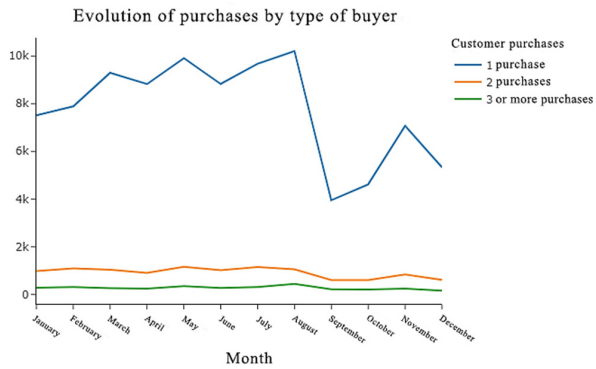

Figure 6: Customers repurchases.

5 DISCUSSION

5.1 Extension

Due to the lack of some data in this study, there is a certain deviation between the actual sales situation of Olist and the analysis and prediction. In the absence of valid review content data, it is impossible to pinpoint the reason for the sudden drop in ratings in August 2017. Further customer feedback data needs to be collected and its trend needs to be analyzed to determine whether Olist's products and services were incorrectly provided, or whether competitors had produced a large number of malicious low ratings.

For sales, two years' data was not enough to make accurate predictions. With the growing popularity of online consumption, Olist's sales are likely to grow even more sharply after 2018. To judge whether Olist is competitive among enterprises of the same type at the same time, more data of competing businesses and itself are needed for comparative analysis before more accurate and effective suggestions can be given.

5.2 Recommendation

Considering the operating condition of Olist, we have three recommendations:

(1) To improve the order validation efficiency, Olist should strengthen the contact with cooperative merchants and shorten the time of order processing. It is suggested that Olist introduce a policy that requires general merchants to process orders within 24 hours, and merchants selling customized goods should also process orders within 48 hours, so as to ensure consumers' shopping efficiency.

(2) To handle the problems with product quality, Olist should strengthen its quality control. Olist can also require suppliers to pay customers the double value of the product and a certain amount of penalty to the platform when quality problems emerge.

(3) To attract Regular customers, Olist may collect customers' email and other information to recommend products and promotion activities regularly. Also, Olist should develop different activities according to different themes throughout the year, so as to attract customers to repeat shopping on the platform within a short period of time.

6 CONCLUSION

Olist is a Brazilian e-commerce platform. According to more than 10,000 pieces of public sales data from 2017 to 2018, the marketing status and existing problems of Olist were analyzed in terms of sales, service condition and customers' satisfaction. In this article, we used Python to visualize the data, and we can clearly see the annual sales trend, the distribution of order confirmation time, the change of customer rating, and the distribution of consumption frequency.

First, based on the annual sales trend, we get into the result that the holiday effects such as Thanksgiving Day and Christmas have a significant stimulus to the sales of Olist. The sales trend in the first half of the year was slightly weaker than in the second half due to the lack of effective motivation. Second, the time of order confirmation and delivery efficiency indicates that Olist, as an e-commerce platform, still has problems in cooperation with suppliers. Although the overall order confirmation time is less than 24 hours, there are still some orders that take more than a day to confirm. As a platform, consumers' shopping experience should be more taken into account, and time should be reduced to improve order efficiency. The last point is that customer rating and consumption frequency are inseparable indicators of the two boxes. According to the data, a considerable number of customers are not satisfied with Olist's services or products, and only a small number of consumers are accustomed to shopping on Olist. We suggested that Olist introduce relevant policies to improve the quality control of platform services and suppliers' products. At the same time, increased marketing efforts to promote the penetration of online shopping in Olist consumer life habits.

While writing this thesis, we realized that there are still many deficiencies as follows: (i) The whole analysis of this paper is based on the data of Olist. So, the data analyzed has great limitations. Besides, the data is only limited to one year's sales, which cannot possess enough universal. The resulting data changes are also applicable to Olist and cannot cover all aspects of online sales. (ii) The commercial data analysis based on Olist makes us lack some innovation and it is difficult for us to innovate the data analysis that belongs to our own creation. (iii) The limitation of data makes the number of references we can use is too small and not authoritative enough.

Based on the defects in the above data analysis, we need to improve our analysis framework in future studies. firstly, we could find out reasons for data changes more clearly and enhance the accuracy of data analysis. Then, the data sources we analyze should also include broader aspects to enhance the universality of our data analysis. Last but not least, we should also add some innovative data analysis

methods in the article to help us make an analysis, so that we can draw some more valuable conclusions.

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